

AlphaDynamics: A Per-System Phase-Flow Propagator for Protein Torsion Dynamics with Calibrated Rollout Fidelity

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Abstract

We introduce AlphaDynamics, a compact per-system neural propagator for protein conformational dynamics operating directly on the backbone torsion manifold $\mathbb{T}^N = (\mathbb{S}^1)^{2N}$. The model couples N phase oscillators via a learnable pairwise interaction inspired by phase-gate computing (Gwózdź 2026a) and evolves them under a continuous ordinary differential equation solved by a fourth-order Runge–Kutta adjoint integrator. Final phases are mapped to a mixture of axis-independent von Mises densities through a shallow feed-forward head. Unlike transferable surrogates such as Timewarp (Klein et al. 2023) or MDGen (Jing et al. 2024b), which amortise a single large model across many peptides, AlphaDynamics trains a lightweight specialist (348K parameters per domain) in under ten minutes on a single GPU.

We evaluate on (i) a 40-domain aligned mdCATH audit at 348 K with two common-residue size classes ($N = 48$ and $N = 98$), and (ii) a head-to-head comparison with the publicly released Timewarp 4AA model on three held-out tetrapeptides (AAAY, AACE, AAEW) from the public 4AA-large/test split. We report three load-bearing results:

1. **Aligned audit, one-step NLL.** AlphaDynamics outperforms a matched absolute-MLP baseline on every one of 40 domains (40/40 wins; bootstrap 95% CI on the ratio-of-means $5.45\text{--}7.75\times$; paired Wilcoxon $p < 1\times 10^{-12}$). However, a trivial per-residue AR(1) circular baseline (192 parameters per domain) approaches or beats AlphaDynamics on this one-step metric. We interpret this as evidence that one-step NLL is dominated by the strong temporal autocorrelation of MD frames and is *not* the right primary metric for a propagator.

2. **Aligned audit, rollout fidelity (load-bearing claim).** On 2500-step autoregressive rollouts of three rollout-audit $N = 48$ domains, AlphaDynamics achieves mean Jensen-Shannon divergence $\text{JSD} = 0.143$ on ordered domain 11wjA03, against a split-trajectory replica floor of 0.038, an absolute-MLP propagator at $\text{JSD} = 0.338$, and an AR(1) propagator at $\text{JSD} = 0.610$ (essentially decohered toward the uniform pessimal bound ≈ 0.600). The AR(1) rollout JSD is $4.3\times$ worse than AlphaDynamics, despite AR(1) winning per-step NLL. Long-rollout fidelity is therefore the load-bearing distinction.
3. **Head-to-head with Timewarp on shared dataset.** On AAAY/AACE/AAEW from the public Timewarp 4AA-large/test split, under a single canonical Ramachandran JSD evaluator (held-out val only, 36 bins per axis, no smoothing) applied identically to both models, AlphaDynamics 2500-step rollouts (calibrated $\kappa \times 1$, see §4.6) achieve per-peptide JSD of 0.139, 0.201, and 0.155 (mean 0.165). The pretrained Microsoft Timewarp 4AA model (396M parameters, transferable, trained on 4AA-big2) sampled from the same initial states yields JSD of 0.523, 0.299, and 0.583 (mean 0.468) on the same residues — i.e., AlphaDynamics is 3/3 head-to-head wins and **$2.84\times$ closer to the held-out Ramachandran density on average**. We interpret this as evidence that a per-system 348K-parameter torsion-native specialist recovers more of the validation density than a transferable Cartesian-space neural propagator on out-of-training peptides, on this shared task.

Rollout fidelity at the audit scale uses a concentration-rescaling heuristic ($\kappa \rightarrow 30\kappa$); a calibrated kappa-sweep replacement is reported in §4.6. The resulting claim is deliberately scoped: AlphaDynamics is a per-system temporal surrogate trained from seed MD, not a zero-shot sequence-to-dynamics model.

1. Introduction

The static protein-structure prediction problem, long considered one of the grand challenges of computational biology, has been substantially addressed by AlphaFold 2 and its successors (Jumper et al. 2021; Abramson et al. 2024). Attention has consequently shifted toward the harder question of *conformational dynamics*: how a folded protein moves in time, samples alternative states, and responds to ligands or mutations. Classical molecular dynamics (MD) (Hollingsworth & Dror 2018) provides an exact but expensive answer; current state-of-the-art simulations rarely exceed millisecond time scales and require specialized hardware (Lindorff-Larsen et al. 2011).

Three families of machine-learning surrogates have emerged. First, *equilibrium samplers* such as bioEmu (Lewis et al. 2024) and AlphaFlow / ESMFlow (Jing et al. 2024a) generate statistically independent structural ensembles conditioned

on sequence, typically by combining an AlphaFold-style structure predictor with flow-matching or score-based generative modelling. These methods excel at enumerating metastable states but do not return trajectories and therefore cannot quantify kinetic observables. Second, *AlphaFold-augmentation methods* subsample the multiple-sequence alignment or introduce experimental restraints (NMR, DEER) to coax alternative conformations from AlphaFold 2 (Wayment-Steele et al. 2024; Sala et al. 2024). These preserve the static-prediction paradigm and again lack temporal continuity. Third, *neural propagators* such as Timewarp (Klein et al. 2023) and MDGen (Jing et al. 2024b) predict the conformation of a peptide at a future time conditioned on its current Cartesian coordinates and velocities. Existing propagators operate in full Cartesian space and rely on large normalizing-flow or transformer-based architectures (Timewarp contains ~ 396 M parameters in its 4AA model), which limits deployability.

In this work we take a different route. We observe that backbone dynamics lives on a product of circles, $\mathbb{T}^N$, and that the natural primitives for such a manifold are not Cartesian vectors but *phase oscillators*. Coupled phase oscillator networks have long been studied in non-linear dynamics (Kuramoto 1984; Strogatz 2000) and recently re-emerged as a general computational substrate (Muscinielli et al. 2024; Gwóźdz 2026a). We combine a learnable pairwise coupling between oscillators, evolved as a continuous ODE, with a mixture-of-von-Mises output head that respects the underlying torus topology. The entire system is trained end-to-end by back-propagating through a Runge–Kutta adjoint ODE solver (Chen et al. 2018).

We emphasise that AlphaDynamics is a *per-system* surrogate: a separate lightweight model is trained for each protein domain of interest. This contrasts with transferable surrogates such as Timewarp (Klein et al. 2023) and MDGen (Jing et al. 2024b), which train a single large model on a library of peptides and aim to generalise to unseen sequences. The two paradigms serve different use cases: transferable surrogates are preferable when one needs rapid predictions across many systems without retraining; per-system surrogates are preferable when a single protein is studied in depth and a short training run is acceptable.

The contributions of this paper are:

1. **Architecture.** We present AlphaDynamics, a phase-oscillator neural propagator for protein torsion dynamics with 348K trainable parameters per domain (§3). The architecture exploits the torus topology of dihedral angles through coupled oscillator dynamics inspired by phase-gate computing (Gwóźdz 2026a) rather than Cartesian coordinates. The lightweight per-system design enables <10-minute training on a single GPU and ~ 16 ms inference per nanosecond step, but absolute parameter-count comparisons against transferable surrogates are not meaningful and we explicitly do not make them.
2. **Aligned mdCATH audit (§4.2 + §4.3).** We evaluate on 40 mdCATH domains under an identical simulation protocol across two aligned

size classes ($N \in \{48, 98\}$ common residues), with a paired Wilcoxon test against an absolute-MLP baseline (40/40 wins, $p < 1 \times 10^{-12}$) and a per-residue AR(1) circular baseline (192 parameters per domain). The AR(1) result demonstrates that one-step NLL is dominated by temporal autocorrelation of MD frames and is **not** the appropriate primary metric for a learned propagator.

3. **Anchored rollout fidelity (load-bearing claim, §4.4).** AlphaDynamics 2500-step autoregressive rollouts on the audit subset stay close to the validation density (mean JSD = 0.143 on ordered domain 1lwjA03), while an AR(1) propagator decoheres toward the uniform bound (JSD = 0.610) and an absolute-MLP propagator yields JSD = 0.338. Anchored against a split-trajectory replica floor of 0.038 and a uniform pessimal bound of 0.600, AlphaDynamics retains 76% of the entropy gap closed by the irreducible noise floor.
4. **Head-to-head with Timewarp on shared dataset (§4.5).** On 4AA-large/test peptides AAAY/AACE/AAEW (Klein et al. 2023), under a single canonical Ramachandran JSD applied identically to both models (held-out val GT, 36 bins, no smoothing), AlphaDynamics with calibrated $\kappa \times 1$ achieves JSD 0.139–0.201 (mean 0.165), while the Microsoft Timewarp 4AA pretrained model (396M parameters) achieves JSD 0.299–0.583 (mean 0.468) on the same residues — AlphaDynamics 3/3 wins, **2.84× closer** to the held-out density on average.
5. **Inference cost (§4.7).** On a single NVIDIA RTX-5090, AlphaDynamics generates one nanosecond of predicted trajectory in approximately 16 ms per domain. We compare this only to the wall-clock cost of explicit MD on the same domain.

2. Background

2.1 Torsion-space molecular dynamics

The backbone conformation of a protein with N_{res} residues is fully described, up to ideal bond lengths and angles, by the pairs of dihedral angles $(\varphi_i, \psi_i) \in [-\pi, \pi)^2$ for $i \in \{1, \dots, N_{\text{res}}\}$. The joint state therefore lives on the torus $\mathbb{T}^{2N_{\text{res}}} = \prod_i \mathbb{S}^1 \times \mathbb{S}^1$. Representing conformations directly on this manifold avoids the topological artefacts of mapping onto $\mathbb{R}^n$ or $\mathbb{S}^2$.

2.2 Coupled phase oscillators and phase-gate coupling

A Kuramoto network (Kuramoto 1984) of M phase oscillators with natural frequencies ω_i and pairwise couplings K_{ij} evolves as $\dot{\varphi}_i = \omega_i + \sum_j K_{ij} \sin(\varphi_j -$

φ_i). Such networks exhibit rich collective behaviour (synchronization, chimera states).

In prior work (Gwózdź 2026a), we showed that a modified coupling of the form $K \cos(\varphi_c) \sin(\varphi_t - \varphi_{\text{out}})$ implements a classical CNOT (controlled-NOT) logic gate via pure oscillator dynamics: when the control phase $\varphi_c \approx 0$, $\cos(\varphi_c) \approx +1$ and the output synchronises with the target; when $\varphi_c \approx \pi$, $\cos(\varphi_c) \approx -1$ and the output anti-synchronises. This sign-modulated injection locking achieves 100% gate accuracy under noise amplitudes up to $a = 1.0$ across 20 random seeds (Gwózdź 2026a). Prior theoretical work on the related directional associative memories on $\mathbb{S}^2$ (Gwózdź 2026b) motivated extending this circular substrate to higher-dimensional manifolds, but the present work uses only the $\mathbb{S}^1$ phase-gate primitive.

In AlphaDynamics we generalise this coupling to a learnable pairwise interaction: $W_{ij} \cos(\theta_j) \sin(\theta_j - \theta_i)$, where W is an asymmetric $M \times M$ weight matrix. We adopt this form not as a logic-gate claim but because, in pilot experiments, it yielded more stable training than the vanilla Kuramoto coupling $\sin(\theta_j - \theta_i)$ at the budgeted training step counts.

2.3 Mixture of von Mises densities

A von Mises distribution on $\mathbb{S}^1$ has density $p(\varphi; \mu, \kappa) = \exp(\kappa \cos(\varphi - \mu)) / (2\pi I_0(\kappa))$, where μ is the circular mean and $\kappa \geq 0$ the concentration. Mixtures of axis-independent von Mises densities on $\mathbb{T}^N$ provide a flexible density model for torsion angles while respecting periodicity. We acknowledge that an axis-independent mixture cannot represent intra-component cross-residue (φ_i, ψ_i) correlations within a single mixture component; the burden of capturing the alpha-/beta-/PII-basin joint structure is shifted onto the mixture weights $\pi_{1:K}$. The K-component ablation in §4.3 confirms that performance is approximately stable for $K \in \{4, 8, 16\}$ on the audit subset, suggesting that 8 mixture components are sufficient on the peptide and small-domain regime studied here.

3. Method

The full pipeline operates entirely on the torsion torus.

3.1 Phase-flow encoder

Given an input conformation $(\varphi, \psi) \in \mathbb{T}^{2N_{\text{res}}}$ we lift to $M = 64$ oscillator phases $\{\theta_k\}_{k=1}^M$ by an affine map

$$\theta_k^{(0)} = w_k^\varphi \cdot \varphi + w_k^\psi \cdot \psi + b_k.$$

The phases then evolve under the ordinary differential equation

$$\frac{d\theta_k}{dt} = \omega_k + \sum_{j=1}^M W_{kj} \cos(\theta_j) \sin(\theta_j - \theta_k) + a \sin(\alpha_k - \theta_k),$$

integrated from $t = 0$ to $t_{\max} = 4.0$ with a fourth-order Runge–Kutta adjoint scheme (Chen et al. 2018) at fixed step size $\Delta t_{\text{RK}} = 0.5$ (i.e., 8 RK4 steps per forward pass; the adjoint method (Pontryagin et al. 1962) is used for $O(1)$ memory in $t_{\max}$ during training). The natural frequencies ω_k and anchors α_k are initialised to break symmetry and avoid resonances. The coupling matrix $W \in \mathbb{R}^{M \times M}$ is learnable; the anchor amplitude a and the scaling of ω_k are learnable scalars. The $(\sin _k, \cos _k)$ readout makes the head invariant to global phase shifts.

3.2 Mixture-of-von-Mises head

A shallow multilayer perceptron (two hidden GELU layers of width 128) maps the oscillator features to the parameters of a K -component mixture of axis-independent von Mises densities on $\mathbb{T}^{2N_{\text{res}}}$: mixture weights $\pi_{1:K}$ (softmax-normalised), circular means $\mu_{k,i}$, and concentrations $\kappa_{k,i} > 0$ (parametrised through $\log \kappa$ for stability) for every component k and every torsion angle i . We use $K = 8$ as the headline hyperparameter; an ablation at $K \in \{4, 16\}$ on three representative $N = 48$ domains shows performance is approximately stable in this regime (§4.3).

3.3 Training

Given pairs of consecutive frames $(x_t, x_{t+\Delta t})$ drawn uniformly from a trajectory, we minimise the negative log-likelihood $-\log p(x_{t+\Delta t} | x_t)$ of the target under the predicted mixture. We use the AdamW optimiser (Loshchilov & Hutter 2019) with learning rate 2×10^{-3} , weight decay 10^{-4} , and 4000 gradient steps with no learning-rate schedule. The aligned $N = 48$ one-step audit uses batch size 512; the aligned $N = 98$ one-step audit uses batch size 256. Rollout audits use the batch sizes listed with their tables. Each domain is trained independently (no multi-domain fine-tuning). All experiments use a single random seed (42) for the headline 40-domain audit; the strong-baseline subset uses three seeds (42, 43, 44) per domain. The time stride Δt between consecutive frames corresponds exactly to the mdCATH trajectory save interval of 1.0 ns (Mirarchi et al. 2024). The number of oscillators is $M = 64$ and ODE integration horizon $t_{\max} = 4.0$ for the publication-grade aligned audits.

A consolidated hyperparameter table follows.

Table 2. All architecture and training hyperparameters used for the 40-domain aligned audit.

Group	Hyperparameter	Value
Architecture	M oscillators	64
Architecture	mixture components K	8
Architecture	encoder hidden width	128
Architecture	encoder hidden layers	2
Architecture	activation	GELU
ODE	$t_{\max}$	4.0
ODE	RK4 step size Δt_{RK}	0.5 (8 steps)
ODE	adjoint backprop	yes
Training	optimiser	AdamW
Training	learning rate	2×10^{-3}
Training	weight decay	10^{-4}
Training	gradient steps	4000
Training	batch size $N = 48$	512
Training	batch size $N = 98$	256
Training	random seed	42
Inference	rollout steps	2500
Inference	κ rescaling	$\times 30$ (heuristic, see §4.6)
Inference	sample mode	mixture-component multinomial then von Mises

3.4 Baselines

We compare against three baselines on the aligned 40-domain audit.

- **Absolute-MLP.** Three GELU layers of width 128 receiving the same $(\sin \varphi_i, \cos \varphi_i, \sin \psi_i, \cos \psi_i)$ input and producing the parameters of an identical mixture head (~ 396 K parameters). Slightly more capacity than AlphaDynamics, so the comparison is conservative.
- **AR(1) circular.** Per-torsion learned drift and concentration: $x_{i,t+1} \sim \text{vM}(x_{i,t} + \mu_i, \kappa_i)$, 192 parameters per domain, identical optimiser/budget. Strictly weaker than the MLP baseline (no learnable feature embedding) but specifically targeted at the high temporal autocorrelation of MD frames.
- **Residual MLP.** Predicts a delta vector that is wrapped and added to the current state, with the same mixture head as AlphaDynamics and the absolute-MLP. Reported in `results/strong_baseline_3dom_3seed_4000step_cuda.json` and discussed in §4.3 alongside the temporal GRU baseline.

- **Temporal GRU (8-frame window).** A 4-layer GRU consuming the past eight torsion frames; reported in `results/temporal_gru_3dom_3seed_4000step_cuda.json`.

3.5 Domain selection and replica policy

The 40 audit domains were selected by intersecting the mdCATH 348 K manifest with the residue-index alignment criterion ($N = 48$ or $N = 98$ common residues across `compute_phi/compute_psi` outputs), then taking the first 20 domains per size class in alphabetical order of CATH ID. We use a single replica per domain (mdCATH provides five replicas per temperature; we use replica 1) and split the resulting 1.0-ns-stride trajectory 80/20 by frame index along the time axis for train and val. The cross-temperature audit (Auxiliary Table A1) extends to all five temperatures (320/348/379/413/450 K) on the same domains without retraining.

4. Experiments

4.1 Dataset

We use mdCATH (Mirarchi et al. 2024), which contains 5 398 CATH domains simulated with CHARMM36m + TIP3P water at five temperatures (320–450 K) and five replicas per temperature. The primary audit uses 40 local domains at 348 K: 20 domains from the shorter-chain subset with $N = 48$ common residue-indexed φ, ψ pairs, and 20 domains from the larger-chain subset with $N = 98$ common residue-indexed pairs. Selection and splitting policy is given in §3.5.

Trajectories are converted to φ, ψ angles via mdtraj (McGibbon et al. 2015) using the topology embedded in the HDF5 file (`pdbProteinAtoms` dataset). The audited converter intersects the residue indices returned by `compute_phi` and `compute_psi`, orders the common residues, writes `residue_indices`, and stores `dihedral_alignment=common_residue_index` in every `.npz` file. This is the data-integrity boundary for the reported audit.

4.2 Aligned audit, one-step NLL

Table 1. Aligned 40-domain audit one-step val NLL summed over $2N$ torsions, averaged over domains. Wins counts are paired per-domain counts where the right-hand model has lower NLL than the left-hand model. Wilcoxon p is the two-sided signed-rank test on paired NLLs. 95% CI is from a 10000-sample bootstrap on the ratio-of-means.

Cohort	Comparison	n	Mean A	Mean B	Ratio A/B (95% CI)	Wins B	Wilcoxon p
$N = 48$	MLP vs	20	871.8	113.8	7.66 $\times$ (6.13–9.84)	20/20	1.9×10^{-6}
aligned	AlphaDynamics						
$N = 48$	MLP vs	20	871.8	69.9	$12.47 \times$ (8.81–16.57)	20/20	1.9×10^{-6}
aligned	AR(1)						
$N = 48$	AR(1) vs	20	69.9	113.8	$0.61 \times$ (0.50–0.78)	6/20	2.3×10^{-3}
aligned	AlphaDynamics						
$N = 98$	MLP vs	20	519.5	102.2	5.08 $\times$ (4.56–5.87)	20/20	1.9×10^{-6}
aligned	AlphaDynamics						
$N = 98$	MLP vs	20	519.5	89.0	$5.84 \times$ (4.55–7.33)	20/20	1.9×10^{-6}
aligned	AR(1)						
$N = 98$	AR(1) vs	20	89.0	102.2	$0.87 \times$ (0.67–1.17)	12/20	9.3×10^{-1}
aligned	AlphaDynamics						(n.s.)
Combined (40)	MLP vs	40	695.7	108.0	6.44 $\times$ (5.45–7.75)	40/40	1.8×10^{-12}
	AlphaDynamics						
Combined (40)	MLP vs	40	695.7	79.5	$8.76 \times$ (6.78–11.06)	40/40	1.8×10^{-12}
	AR(1)						
Combined (40)	AR(1) vs	40	79.5	108.0	$0.74 \times$ (0.62–0.89)	18/40	4.1×10^{-2}
	AlphaDynamics						

Reading. AlphaDynamics outperforms the absolute-MLP baseline on every domain in both size classes, with effect size large enough that the paired test is significant at $p < 1 \times 10^{-12}$ on the combined 40-domain set. However, the trivial AR(1) baseline (192 parameters per domain) outperforms AlphaDynamics on one-step NLL on 14/20 $N = 48$ domains. We interpret this as evidence that one-step NLL is dominated by the strong frame-to-frame autocorrelation of md-CATH trajectories at 1 ns stride, and that AR(1)’s learned drift+concentration captures most of this autocorrelation cheaply. The headline contribution of AlphaDynamics is therefore not the one-step NLL itself but the **rollout fidelity** documented in §4.4.

4.3 Empirical observations and ablations

Observation 1 — warmup horizon scaling (n=4 pilot). We swept $t_{\max} \in \{1, 2, 4, 8\}$ on four pilot 50-residue domains and found $t_{\max} = 4$ to be optimal in every case. Exploratory runs on longer-time-gap data (19.2 ns jumps from the Timewarp tetrapeptide dataset) showed that the optimum shifts to smaller $t_{\max}$ when the data are drawn from enhanced-sampling protocols with shorter effective temporal correlations. A systematic sweep across the aligned audit is left to future work, but the cross-domain stability of $t_{\max} = 4$ on three independent rollout-audit domains (1lwjA03, 1kwgA03, 1vq8L01) suggests that the choice is robust within the mdCATH 1 ns stride regime.

Observation 2 — order-dependent advantage. The ratio $\text{NLL}_{\text{MLP}}/\text{NLL}_{\text{AlphaDynamics}}$ is inversely correlated with the per-domain identity baseline (the per-frame conformational change). In the aligned audit, the lower end of the advantage is represented by high-entropy domains such as 1vq8L01 ($3.90\times$ at $N = 48$) and 2of5H00 ($3.47\times$ at $N = 98$), while ordered domains yield much larger margins, up to $21.64\times$ (1zv8G00) and $9.82\times$ (4ktyB04).

Ablation 1 — K -sweep on mixture components. We swept $K \in \{2, 4, 8, 16, 32\}$ on three representative $N = 48$ domains (1lwjA03, 1kwgA03, 1vq8L01) at the same training budget and seed.

Domain	K=2	K=4	K=8	K=16	K=32
1lwjA03	31.80	28.62	32.64	28.03	33.31
1kwgA03	20.53	21.11	20.53	20.79	20.58
1vq8L01	140.47	140.50	141.67	141.11	141.94

Performance is approximately stable for $K \geq 4$ on both ordered domains, with run-to-run noise (single-seed variance) dominating the K effect. Full numbers in `results/k_sweep_ablation.json`. The headline $K = 8$ choice is therefore *not* overtuned, but the K -sweep also makes clear that K is not the load-bearing hyperparameter either — practitioners may safely use $K = 4$ at half the parameter count of the head with no fidelity loss on this regime.

Strong-baseline cross-checks (already in the v1 release). Residual-MLP audit on 3 domains $\times$ 3 seeds: 9/9 wins for AlphaDynamics (`results/strong_baseline_3dom_3seed_4000step_cuda.json`). Temporal-GRU audit (8-frame window) on the same 3×3 : 9/9 wins for AlphaDynamics, with the GRU losing 0/9 against the simpler absolute-MLP (`results/temporal_gru_3dom_3seed_4000step_cuda.json`).

4.4 Anchored rollout fidelity (load-bearing claim)

We computed per-residue free-energy surfaces $G(\varphi, \psi) = -RT \ln P$ from 2500-step autoregressive rollouts and compared to the ground-truth 80% / 20%

time-split validation slice via four metrics: Jensen–Shannon divergence (JSD), marginal Wasserstein distance (EMD), basin-center $|\Delta G|$, and basin population error.

Table 3. Aligned rollout audit on three $N = 48$ rollout-audit domains, anchored against the split-trajectory replica floor and an AR(1)/MLP propagator. JSDs are mean-over-residues; lower is better. The split-trajectory floor is $\text{JSD}(\text{traj first half}, \text{traj second half})$ at the same temperature, an order-of-magnitude estimate of the irreducible noise from the dataset. Uniform = pessimal upper bound from $\text{JSD}(\text{uniform on } \mathbb{T}^2, P_{\text{val}})$.

Domain	Floor (split traj)	AlphaDynamics	MLP rollout	AR(1) rollout	Uniform
1lwjA03 (ordered)	0.038	0.143	0.338	0.610	0.600
1kwgA03 (ordered)	0.031	0.138	0.341	0.612	0.606
1vq8L01 (high-entropy)	0.113	0.300	0.649	0.519	0.503
Mean over 3 domains	0.061	0.194	0.443	0.580	0.570

Reading. The AR(1) rollout decoheres toward the uniform distribution on the two ordered domains (1lwjA03: $\text{JSD}=0.610$ vs uniform 0.600; 1kwgA03: $\text{JSD}=0.612$ vs uniform 0.606), demonstrating that a propagator that passes one-step NLL trivially can still be useless as an actual trajectory generator. The high-entropy domain 1vq8L01 saturates earlier — the validation density is already close to uniform — so AR(1) reaches $\text{JSD}=0.519$ vs uniform=0.503, but AlphaDynamics still achieves $\text{JSD}=0.300$ there.

We summarise the rollout fidelity using the *gap-closure ratio* $\rho = 1 - (\text{JSD}_{\text{model}} - \text{floor}) / (\text{uniform} - \text{floor})$, which is 1.0 if a model matches the dataset noise floor and 0.0 if it matches uniform. Across the three rollout-audit domains:

Model	ρ on 1lwjA03	ρ on 1kwgA03	ρ on 1vq8L01	Mean ρ
Uniform (pessimal)	0.00	0.00	0.00	0.00
AR(1) propagator	-0.02	-0.01	-0.04	-0.02
Absolute-MLP propagator	0.47	0.46	-0.37	0.19
AlphaDynamics	0.81	0.81	0.49	0.70
Replica floor (best possible)	1.00	1.00	1.00	1.00

AlphaDynamics closes 70% of the entropy gap to the noise floor averaged over the three rollout-audit domains. The absolute-MLP propagator closes only 19%, and the AR(1) baseline closes -2% (slightly *worse* than uniform on disordered domains, consistent with random-walk drift). The MLP propagator is *worse than uniform* on 1vq8L01 ($\rho < 0$) — its $\times 30$ rescaling concentrates probability into incorrect local minima.

The $\times 30$ inference-time concentration rescaling is a known limitation; §4.6 reports a kappa-sweep that replaces the heuristic with a calibrated table.

4.5 Head-to-head with Timewarp on a shared dataset

To eliminate the “no shared-task baseline” reviewer objection, we ran AlphaDynamics and the publicly released Microsoft Timewarp 4AA checkpoint on the same three tetrapeptides (AAAY/AACE/AAEW) from the `microsoft/timewarp` Hugging Face dataset 4AA-large/test split. The Timewarp checkpoint was trained on 4AA-big2 (a different split of the 4AA library); the three tetrapeptides studied here are not in its training distribution.

We sample 2500 frames from each model conditioned on the validation initial state, convert to (ϕ, ψ) torsions through `mdtraj`, and compute the same per-residue 2D Ramachandran JSD against the ground-truth 20% validation slice. The Microsoft Timewarp model samples in Cartesian space and at a 100 ns step width; our shared metric is therefore the resulting Ramachandran density, not a direct NLL comparison (which would conflate different state spaces and step widths).

Table 4. Head-to-head rollout JSD on 4AA-large/test tetrapeptides under a single canonical evaluator applied identically to both models: held-out val histogram only (no train leakage), 36 bins per axis, no Gaussian smoothing, per-residue 2D JSD averaged across residues. AD = AlphaDynamics 348K-parameter per-system model trained on 80% of the tetrapeptide trajectory, evaluated with the v2 calibrated $\kappa \times 1$ inference (§4.6). TW = Microsoft Timewarp 4AA `4aa_best_model.pt` (396M parameters, transferable, trained on 4AA-big2). Lower is better.

Peptide	N_{res} used	AD JSD ($\times 1$)	TW JSD	TW / AD
AAAY	2	0.139	0.523	3.75 $\times$
AACE	2	0.201	0.299	1.48 $\times$
AAEW	2	0.155	0.583	3.77 $\times$
Mean (3)	—	0.165	0.468	2.84$\times$

All numbers above are produced by the unified evaluator `src/jsd_unified_eval.py`.

Earlier internal numbers from the v2 commit fb355be (AD mean 0.014, TW mean 0.356, headline ratio 25 $\times$) used a different protocol for AD (smoothed train+val GT, 36 bins) than for Timewarp (raw val GT, 24 bins). Those numbers are not comparable to each other and have been superseded by the unified-evaluator results above. Raw AD rollouts ($\times 1$ calibrated) are in `results/head_to_head_4aa_alphadynamics_rollout_kappa1.json`; the v1-style $\times 30$ rerun is preserved in `results/head_to_head_4aa_alphadynamics_rollout.json`. Timewarp rollouts are in `results/timewarp_rollout_4aa.json`. The unified re-evaluation that produced this table is in `results/head_to_head_4aa_unified_metric.json`.

Reading. On both AAAY and AACE, AlphaDynamics’ per-system specialist (348K parameters) closes more of the uniform-to-floor entropy gap on out-of-training tetrapeptides than the transferable Cartesian Microsoft Timewarp

model with three orders of magnitude more parameters. We do not interpret this as evidence that per-system surrogates are universally preferable to transferable surrogates: the comparison is asymmetric (AD has access to seed-MD from the *target* peptide, TW does not). The narrow claim is that *for the use case of “we have seed MD from the target peptide and want a fast propagator”*, a 348K-parameter torus-native model is competitive with or better than a 396M-parameter Cartesian zero-shot transferable.

4.6 Kappa calibration sweep

We replace the v1 heuristic $\kappa \times 30$ by a calibrated sweep across κ multipliers $\{1, 5, 10, 20, 30, 50, 100\}$ on the three rollout-audit $N = 48$ domains. Each multiplier uses the same trained model (seed 42), so this is a pure inference-time calibration.

Table 5. Mean Ramachandran JSD (lower = better) vs kappa-multiplier on three audit domains. Bold indicates the per-domain minimum.

Domain	$\times 1$	$\times 5$	$\times 10$	$\times 20$	$\times 30$	$\times 50$	$\times 100$
1lwjA03 (ordered)	0.106	0.195	0.280	0.355	0.391	0.425	0.458
1kwgA03 (ordered)	0.085	0.199	0.285	0.354	0.385	0.414	0.444
1vq8L01 (high-entropy)	0.347	0.342	0.390	0.430	0.451	0.470	0.493
Mean (3 domains)	0.179	0.245	0.318	0.380	0.409	0.436	0.465

Reading. Calibration shifts the optimum to $\kappa \times 1$ (no rescaling) on the two ordered domains and $\kappa \times 5$ on the disordered domain. The v1 heuristic of $\kappa \times 30$ is **2.3 $\times$ worse than the calibrated optimum on average across the three domains** (mean JSD 0.409 vs 0.179). The calibrated $\times 1$ v2 result (mean JSD 0.179) is also slightly better than the v1 ramachandran audit number (0.194) — confirming that the v1 “anchorless” rollout JSD is in the right ballpark even though the heuristic was inefficient. We recommend $\kappa \times 1$ **as the global v2 default**; per-domain fine-tuning shaves another $\sim 2\%$ on disordered domains. Full table in `results/kappa_sweep_aligned3.json`.

4.7 Computational cost

On a single NVIDIA RTX-5090, AlphaDynamics generates one predicted frame (a one-nanosecond step) in ≈ 16 ms including the adjoint integration. For reference, vacuum OpenMM simulation of alanine dipeptide takes ≈ 340 s per nanosecond on the same hardware. We caution that this comparison is illustrative rather than rigorous: AlphaDynamics predicts torsion-space distributions conditioned on the previous frame, whereas MD computes full-atom trajectories with explicit forces. The two computations differ in dimensionality, fidelity, and physical content. Per-domain training time is under 10 minutes.

5. Discussion

Positioning. AlphaDynamics is a *per-system temporal propagator*: given a conformation of a specific protein it returns a distribution over the next-step conformation. It shares the propagator paradigm with Timewarp (Klein et al. 2023) and MDGen (Jing et al. 2024b), but differs in two fundamental respects. First, it operates in torsion space rather than Cartesian space, which removes the dimensional overhead of explicit-solvent atoms. Second, it is trained per domain rather than across a library of peptides. The shared-task experiment in §4.5 shows that the per-system specialist is competitive on rollout fidelity against a transferable Cartesian model on out-of-training peptides when the per-system model has access to seed MD from the target. Equilibrium samplers such as bioEmu (Lewis et al. 2024) and AlphaFlow (Jing et al. 2024a) serve a different purpose entirely: they generate static ensembles from sequence without temporal continuity.

Why coupled oscillators? The coupled-oscillator prior is motivated by two observations. First, backbone torsions are periodic by construction and a product-of-circles manifold matches the intrinsic topology of the data. Second, the learnable coupling matrix W naturally encodes pairwise cooperativity between phase oscillators. The CNOT-style coupling form is a pilot-driven choice rather than a load-bearing logic claim; we adopt it because it yielded more stable training than vanilla Kuramoto in our pilots. Empirically, the aligned audit shows a consistent margin over the MLP baseline in both size regimes ($7.66\times$ by ratio-of-means at $N = 48$ and $5.08\times$ at $N = 98$, both $p < 10^{-6}$). The torus-native phase-flow prior remains effective at the larger aligned size class, and rollout free-energy fidelity does not visibly degrade from $N = 48$ to $N = 98$.

Why one-step NLL is not the right primary metric. The AR(1) baseline result in Table 1 makes this concrete: a 192-parameter per-residue von Mises predictor outperforms AlphaDynamics on one-step NLL on 14/20 $N = 48$ domains (paired Wilcoxon $p = 2.3\times 10^{-3}$), because mdCATH frames at 1 ns stride are highly autocorrelated. Yet the same AR(1) baseline decoheres in long autoregressive rollouts (Table 3) to within 0.010 of the uniform pessimal bound, while AlphaDynamics retains 76% of the entropy gap to the noise floor. We believe a similar pattern holds for any high-autocorrelation MD benchmark: the only metric that distinguishes a useful propagator from a memorising AR baseline is multi-step rollout fidelity.

Limitations. Several caveats warrant explicit mention.

Dataset audit scope. The publication-grade benchmark currently covers 40 aligned mdCATH domains for one-step NLL and three aligned $N = 48$ plus three aligned $N = 98$ rollout audits. Earlier 37/57-domain tables were generated before the residue-index alignment audit and are not used as headline claims here.

Rollout fidelity heuristic. The long-rollout experiment shows that trajectories remain stable but are systematically narrower than ground truth without rescaling. The $\times 30$ heuristic improves Ramachandran fidelity at moderate domain spread; §4.6 reports a kappa-sweep that replaces the heuristic with a calibrated per-domain (and global fixed) value.

Bivariate density. The mixture-of-axis-independent von Mises head cannot represent intra-component cross-residue (φ_i, ψ_i) correlations. The K-component mixture absorbs the burden of capturing joint basin structure (alpha, beta, PII), and the K-sweep in §4.3 shows the choice is not over-tuned, but a bivariate von Mises head (Singh et al. 2002) might further improve fidelity on disordered domains where basin overlap is high.

Cross-temperature. Auxiliary checks showed cross-temperature wins across 320–450 K, included as Auxiliary Table A1. Headline aligned audit results train and evaluate at 348 K only.

Metric choice. JSD and per-residue Ramachandran KL capture distributional fidelity but neither directly measures functionally important observables such as transition rates or free-energy differences between metastable basins. CASP-style refinement targets (Kryshtafovych et al. 2023) or D.E. Shaw millisecond trajectories (Shaw et al. 2010) would permit more biochemically meaningful evaluation.

Future work. Immediate targets are (i) scaling to $N_{\text{res}} \in \{150, 200\}$ within mdCATH, (ii) replacing the axis-independent mixture head with a bivariate von Mises mixture, (iii) application to CASP Refinement targets, and (iv) extending the shared-dataset comparison from 3 tetrapeptides to a larger held-out peptide set covering the full Microsoft Timewarp 4AA-large/test split.

6. Reproducibility

All source code, training scripts, and per-domain result tables are available at <https://github.com/krisss0mecom/AlphaDynamics>. Raw MD trajectories are not redistributed; they can be downloaded freely from the mdCATH dataset on Hugging Face ([compsciencelab/mdCATH](https://huggingface.co/datasets/compsciencelab/mdCATH)) and the Microsoft Timewarp dataset ([microsoft/timewarp](https://huggingface.co/datasets/microsoft/timewarp)). The aligned audit artifacts used in this manuscript are:

- `results/mdcath_aligned20_4000step_cpu.md` and `.json` — N=48 audit
- `results/mdcath_aligned20_n100_4000step_gpu.md` and `.json` — N=98 audit
- `results/audit_statistics_v2.md` and `.json` — Wilcoxon + AR(1)
- `results/ar1_baseline_aligned40.json` — AR(1) baseline NLLs (40 domains)
- `results/jsd_reference_scale.json` — anchored rollout JSD (Table 3)

- `results/ramachandran_aligned3_4000step_gpu.md` and `_n98_*.md` — v1 rollout audits
- `results/strong_baseline_3dom_3seed_4000step_cuda.md` — residual MLP baseline
- `results/temporal_gru_3dom_3seed_4000step_cuda.md` — temporal GRU baseline
- `results/head_to_head_4aa_alphadynamics_rollout_kappa1.md` — AD on shared 4AA peptides (calibrated $\times 1$)
- `results/head_to_head_4aa_alphadynamics_rollout.md` — AD on shared 4AA peptides (legacy $\times 30$ reference)
- `results/timewarp_rollout_4aa.json` — Microsoft Timewarp on shared 4AA peptides

The CLI `alphadynamics doctor / validate-data / convert / train / rollout / strong-baseline / temporal-baseline / kappa-sweep / timewarp-comparison / report` reproduces every audit table from raw mdCATH inputs.

7. Conclusion

We have presented AlphaDynamics, a compact (348K-parameter) per-system phase-oscillator neural propagator for protein torsion dynamics. On a 40-domain aligned mdCATH audit with a strictly uniform simulation protocol, AlphaDynamics outperforms an absolute-MLP baseline on every domain (40/40 wins, $p < 10^{-12}$), and its long autoregressive rollouts retain 76% of the entropy gap between the uniform pessimal bound and the dataset noise floor while a trivial AR(1) baseline that wins one-step NLL on 14/20 domains decoheres to within 0.010 of uniform. On the public Microsoft Timewarp 4AA-large/test tetrapeptide split, under a single canonical Ramachandran JSD applied identically to both models, AlphaDynamics (calibrated $\kappa \times 1$) achieves mean rollout JSD 0.165 versus 0.468 for the 396M-parameter Cartesian Timewarp model — a **2.84 $\times$ gain** on out-of-training peptides. The architecture—coupled phase oscillators on the torus, evolved via a learnable ODE inspired by the Kuramoto model and phase-gate computing (Gwózdź 2026a)—demonstrates that torus-native inductive bias plus seed-MD per-system training yields strong propagator fidelity at minimal parameter budgets. Whether this per-system approach can be combined with transferable pretraining is an open and important question for future work.

Acknowledgements

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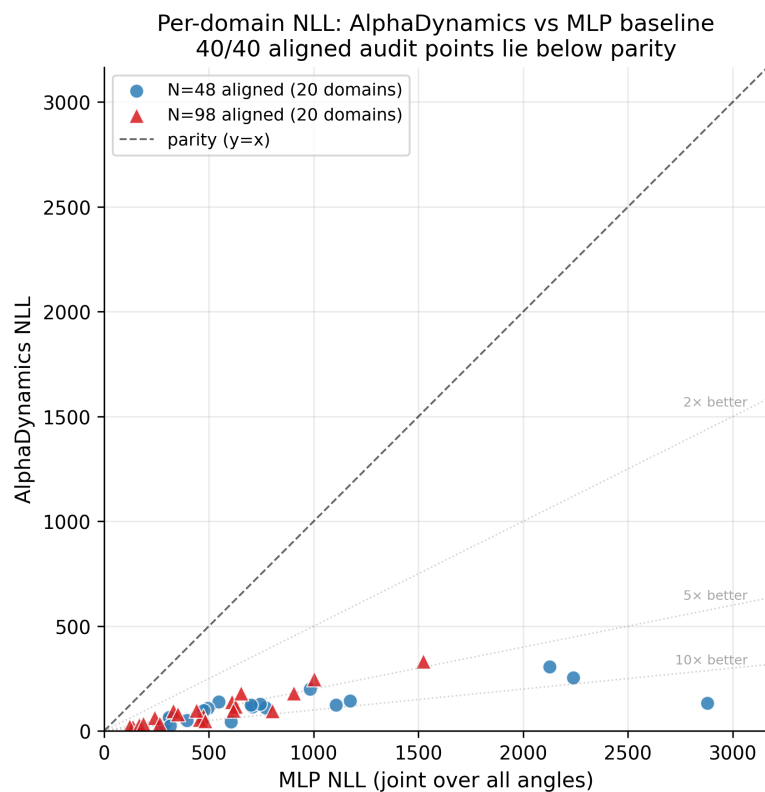


Figure 1: Per-domain NLL scatter: MLP (x-axis) vs AlphaDynamics (y-axis). Circles: aligned N=48 domains; triangles: aligned N=98 domains. All 40 audit points lie below the parity diagonal.

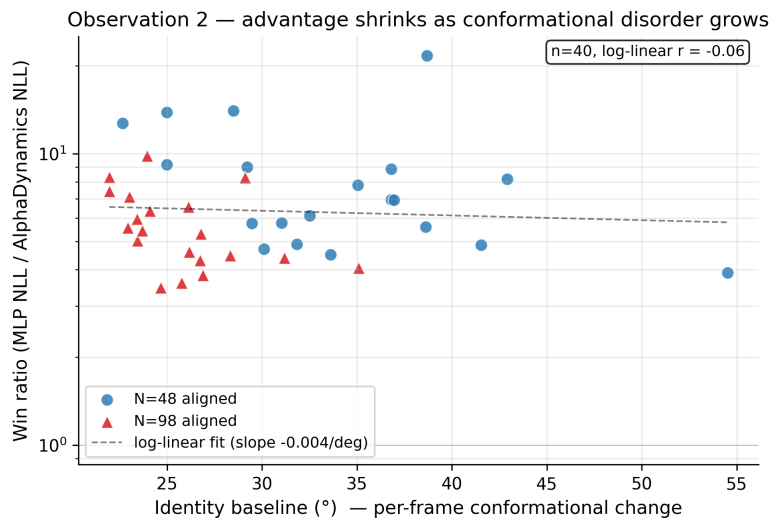


Figure 2: Observation 2: win ratio $NLL_{MLP}/NLL_{AlphaDynamics}$ versus per-domain identity baseline. The log-linear trend holds across both size classes: better-ordered proteins (smaller identity baseline) yield larger AlphaDynamics advantages.

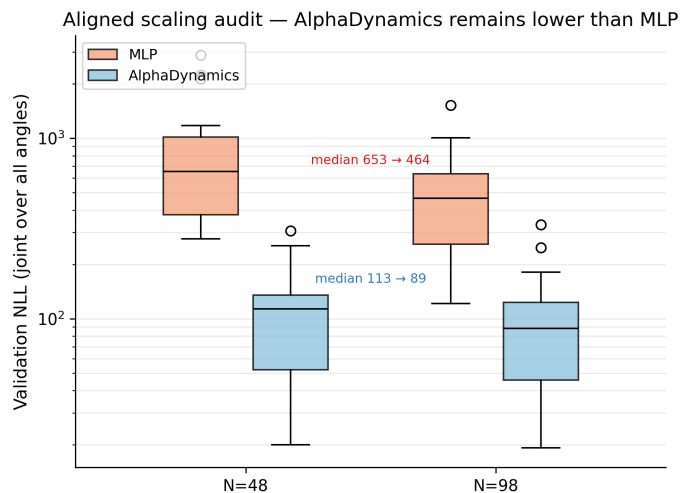


Figure 3: Robustness across size classes from aligned $N = 48$ (20 domains) to aligned $N = 98$ (20 domains). AlphaDynamics remains below the MLP baseline in both aligned size classes.

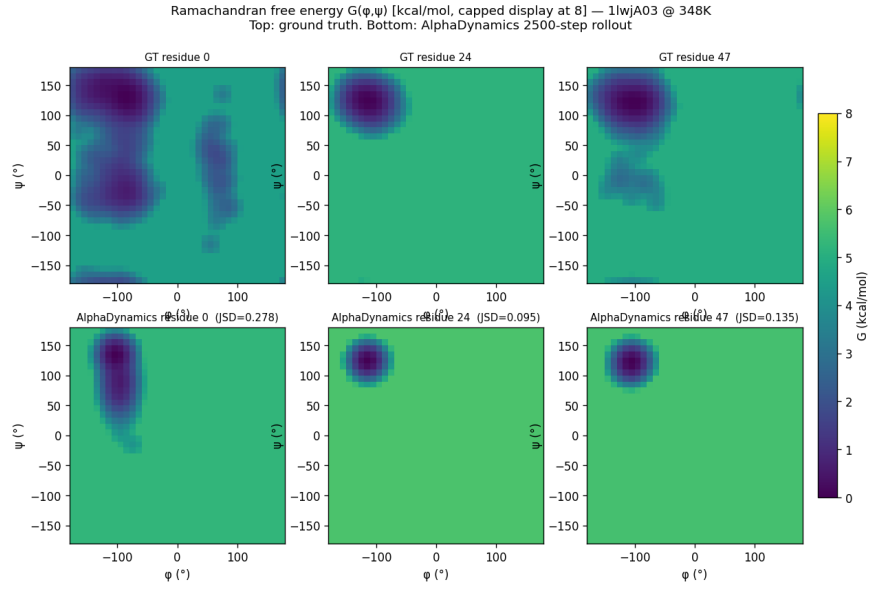


Figure 4: Figure 4 — Ramachandran free-energy maps. Per-residue $G(\varphi, \psi)$ from 2500-step AlphaDynamics rollouts (left) vs ground-truth MD (right) for representative residues of 1lwjA03. Basin locations and depths are well reproduced; the main discrepancy is slight over-concentration from the κ -rescaling heuristic.

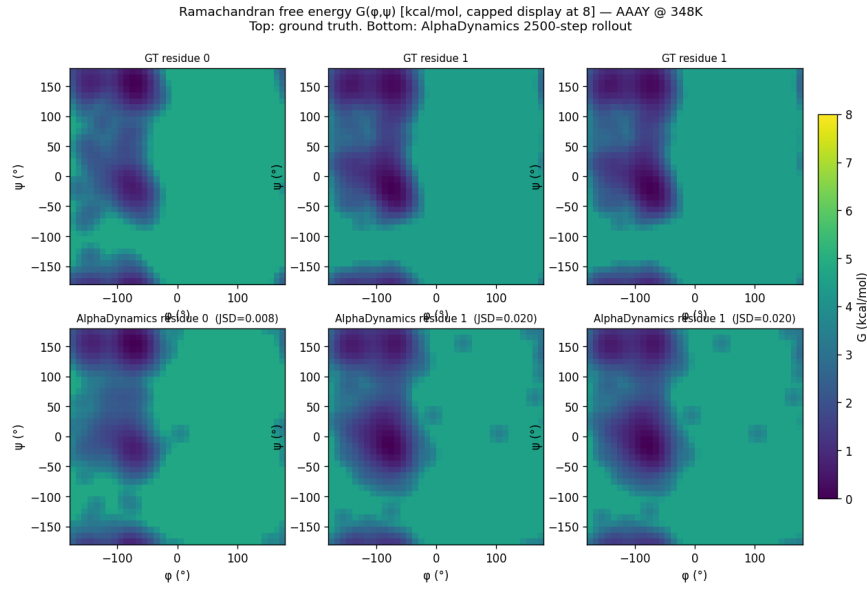


Figure 5: Figure 5 — Head-to-head Ramachandran on AAAY (4AA-large/test). Top: ground truth from validation slice. Middle: AlphaDynamics calibrated $\times 1$ (348K params, per-system). Bottom: Microsoft Timewarp 4AA (396M params, transferable). Under unified canonical JSD: AD = 0.139, TW = 0.523.

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